Rui Zhang

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RESEARCH PROFILE

MEng graduate in Environmental Geological Engineering specialising in earthquake-triggered submarine landslides and marine geohazards. My work combines mechanics-based modelling, GPU computing, soil testing and industry-linked geological-risk assessment. I am first author of a 2026 paper in Computers and Geotechnics, a Q1 journal with a 2025 JCR Impact Factor of 7.1.

EDUCATION

Ocean University of China Sep 2023 - Jun 2026

MEng in Environmental Geological Engineering

- GPA: 3.76/4.0
- Research focus: earthquake-triggered submarine landslides and marine geohazards
- Supervisor: Professor Dong Wang

Ocean University of China Sep 2019 - Jun 2023

BEng in Environmental Engineering

- GPA: 3.77/4.0; Honors Bachelor's Degree, 2023

PUBLICATIONS

Zhang, R., Wang, D. & Zhang, W. (2026). *A depth-averaged method modeling three-dimensional submarine landslides caused by earthquakes.* Computers and Geotechnics, 194, 108012. https://doi.org/10.1016/j.compgeo.2026.108012

RESEARCH EXPERIENCE

National Natural Science Foundation of China Graduate research

Distinguished Young Scholars Project: mechanisms of shallow submarine landslides in the northern South China Sea

- Derived and implemented a depth-averaged method for the full co-seismic evolution of three-dimensional submarine landslides. It includes three-component seismic loading, strain-softening soil behaviour and terrain curvature.
- Built the solver in MATLAB and parallelised it on a GPU, cutting computational cost to about 10% of the CPU version.
- Led the numerical implementation, verification, case-study analysis, visualisation and first-author manuscript preparation.

CNOOC industry collaboration Industry Collaboration Project

Evolution and mechanisms of submarine landslides on a continental slope in the South China Sea

- Worked with the South China Sea oil-and-gas development project group on geological-risk assessment.
- Ran dynamic simulations across earthquake-intensity scenarios and analysed maximum impact velocity, run-out distance, deposition extent and deposit thickness.

Asia Oceania Geosciences Society Annual Meeting Poster presentation

A feasibility study of co-ordinated emissions reduction and pollution control under carbon neutrality

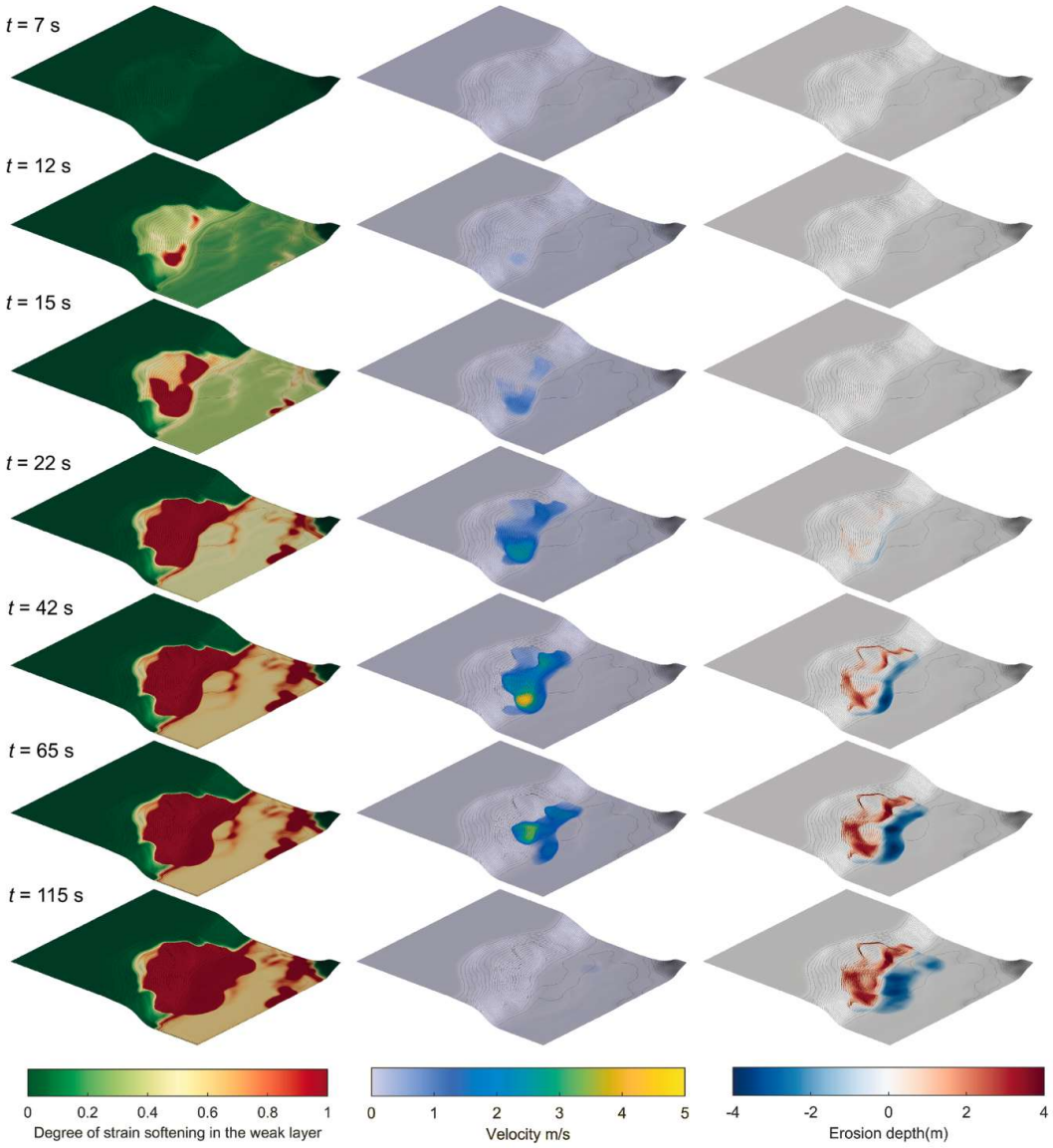
- Presented and discussed the poster in English, explaining the research question, method and results to researchers from several disciplines.

Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences Two-month secondment

- Developed an algorithm to reconstruct palaeo-landslide morphology across the complex terrain of the South China Sea.

SELECTED RESEARCH SUMMARY

A depth-averaged method was developed to simulate the full evolution of an earthquake-triggered submarine landslide, from initiation and shear-band propagation to post-failure runout. Recorded acceleration histories in three directions were applied, with strain softening and three-dimensional terrain curvature taken into account. The method was verified against three benchmark cases and then applied to the St Niklausen slide in Lake Lucerne. The predicted scar and runout were consistent with field observations. The relationship between peak ground acceleration and the area affected by sliding was examined through a parameter study.



Numerical modelling of the St. Niklausen slide evolution under seismic loading. Zhang et al. (2026).

COMPUTATIONAL AND TECHNICAL SKILLS

Programming and computing: MATLAB, including five years of numerical modelling, GPU acceleration, automation and visualisation; Python; Linux; spectral-element simulation of submarine seismic-wave propagation.

Geotechnical software: Proficient in Abaqus, including large-deformation landslide modelling; built a two-dimensional earthquake-induced landslide model in PLAXIS; reproduced published Peridynamics case studies and understand the method's workflow.

Data-informed methods: Studied physics-informed machine-learning workflows and basic principles. I have not yet developed an independent PIML model.

Languages: Mandarin, native; English, fluent, IELTS 7.5; German, beginner.

LABORATORY EXPERIENCE

- Won First Prize in the Geotechnical Experimental Skills Competition at Ocean University of China. Later served twice as a postgraduate judge, helping to design the experimental tasks and marking scheme and assess participants' work.
- Took part in centrifuge physical-model tests of submarine landslides and offshore trials of a free-fall penetrometer.
- Conducted simple-shear and consolidation tests and carried out several triaxial tests using a GDS Triaxial Testing System.

SELECTED AWARDS AND HONOURS

- National Scholarship of China, Ministry of Education of China, 2021 and 2022.
- Outstanding Graduate of Shandong Province, 2023; Honors Bachelor's Degree, Ocean University of China, 2023.
- First-Class Scholarship for Graduates, 2024; First-Class Scholarship for Undergraduates, 2020, 2021 and 2022.